

Q2-01-B

Let

$$f(x) = e^x.$$

1. Find the Taylor polynomial of degree 4 for $f(x)$ about the point $x_0 = 0$.
2. We use this Taylor polynomial to approximate

$$f(-0.1) \quad \text{and} \quad f(0.1).$$

Use Taylor's theorem with remainder to find an error formula for the Taylor polynomial. Give an upper bound for the error in each of the two approximations.